

Response to Reviewers

Paper 1217: Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

Summary

We thank the SPC and the three reviewers for their careful and constructive comments. We were encouraged that the reviewers found the question timely and relevant to ICWSM (SPC, R2, R3) and the cross-platform comparison valuable, particularly because Truth Social and Bluesky remain understudied in comparative research (SPC, R1). We found the feedback very helpful in identifying several places where the original manuscript needed clearer definitions, additional validation, and more cautious interpretation, and we have revised the manuscript substantially in response.

Across these sensitivity analyses, estimates sometimes change substantially in magnitude, but none overturns the central conclusions. Where a check narrows the interpretation of a particular finding, we have added that qualification to the manuscript.

Below we respond first to the seven priorities in the SPC’s roadmap and then to each reviewer’s comments in turn. Reviewer comments are shown in italics and our responses in blue. Throughout, the platform difference means the difference between Truth Social and Bluesky in how cascade breadth or depth grows with cascade size (the platform-by-size interaction in the paper’s regressions), and the share accounted for means how much of that difference disappears once ideological composition and interaction alignment are added to the model. Where a reviewer’s comment is addressed in full under an SPC priority, we summarize the change and refer to that response. Page numbers refer to the revised PDF. All revised and added content in the manuscript is shown in blue; the coloring will be removed in the final version.

Responses to the SPC

- **SPC-1.** *Validate or revise the repost cascade analysis. Provide robustness checks showing how results change under alternative repost reconstruction assumptions, including assumptions about follower-network timing and inferred exposure paths. If this cannot be done convincingly, consider reducing the emphasis on repost cascades or focusing the main contribution on reply cascades.*
 - Thank you for this suggestion, which has made the paper considerably more robust. We added a new appendix (Appendix D, pp. 17-18) that rebuilds all 244,129 repost cascades (4.03 million reposts) under alternative reconstruction rules. Every admissible rule chooses a repost’s parent from the same candidates, the original post plus earlier reposters whom the reposting user follows, so we compared the rule used in the paper with rules that choose the earliest eligible reposter, the most recent one, or a random one (40 draws), each under both readings of the repost list order. For reply cascades, which

are observed directly and need no rule, the platform difference is $+0.170$ for breadth and -0.182 for depth. For repost cascades, the lowest and highest values across all the rules are -0.012 and $+0.031$ for breadth and -0.083 and $+0.028$ for depth (Figure 9, p. 18). Both ranges include zero, so some rules show no platform difference at all, and even the largest repost value is less than a fifth of the reply value for breadth and less than half of it for depth. The rule used in the paper gives the largest repost difference of any rule, and the choice of rule has limited room to matter, since 55 percent of Bluesky reposts and 59 percent of Truth Social reposts have only the original post as a possible parent. The conclusion that reposts scale similarly on the two platforms therefore holds under every rule we examined.

- On follower-network timing: the follower networks are snapshots without edge creation dates, so a follow edge can postdate the repost it is used to explain. Both networks were collected at the same time, so this error applies comparably to the two platforms. To see how much it could matter, we deleted follow edges at random before reconstruction, at rates from 5 to 30 percent. We agree that 30 percent is far more than such a timing gap is likely to produce; we included it as a stress test. At 5 percent the estimates barely move (0.028 for breadth and 0.005 for depth, against 0.031 and 0.004 with no deletion), and across the whole range they stay between 0.004 and 0.031, far below the reply-cascade values. This check is also in Appendix D (p. 17).
- We also removed the description of repost cascades as empirical. They are called reconstructed throughout, and Section 3.2 (p. 4) states how they are built and that the follower network is a post-collection snapshot.
- **SPC-2.** *Clarify cascade definitions. Explicitly state whether nodes are users, posts, or interactions; how repeated users in the same thread are represented; and how these choices affect depth, breadth, and motif analyses.*
 - Thank you for raising this! Nodes are posts. In a reply cascade each node is the root post or a reply, so a user who replies several times contributes one node per reply; in a repost cascade each node is a repost. Ideology is aggregated at the user level, and each node carries the label of the account that wrote it. The submitted version described cascade nodes as users, which did not match the implementation, and Section 3.2 (p. 4) now gives the definition for both cascade types.
 - Because this choice affects depth and breadth, we added a new appendix (Appendix E, p. 18) that recomputes every reply cascade with each user collapsed to a single position, the shallowest depth at which that user appears. The metrics are then computed from these positions alone: size is the number of distinct users, breadth is the largest number of users first appearing at the same depth, and depth is the deepest first appearance. No edges are redrawn. For example, if user a posts, c replies to a, b replies to c, and a replies to b, then a is kept at depth 0, c at depth 1, and b at depth 2, giving a depth of 2, and there is no edge back from a to c. We compared the two representations on the same 42,947 threads, those that still have more than one user after collapsing. The depth difference between platforms persists (-0.184 at post level, -0.132 collapsed) and is again largely accounted for by composition and alignment. The breadth difference does not persist: it falls from 0.160 to -0.032 on the same threads, so part of Truth Social’s greater post-level breadth reflects the same users posting repeatedly within a thread. The paper now states this.
 - Motifs are counted on the post-level trees, so one user can occupy two positions in a motif. To show how often that happens, we added a new appendix (Appendix F, p. 18)

that counts exchanges of the form $A \rightarrow B \rightarrow A$, in which a reply’s author also wrote its grandparent post. Among cascades of depth two or more, 68.8 percent on Bluesky and 70.0 percent on Truth Social contain at least one, and such exchanges account for 20.0 and 15.6 percent of reply edges. Repeat participation of this kind is similarly common on both platforms.

- **SPC-3.** *Strengthen ideology-label validation. Report more detailed validation results, including class-specific and platform-specific performance. Clarify the meaning of the center category and test whether the main findings are robust to plausible ideology-label noise.*
 - Thank you for these suggestions, which, we believe, substantially strengthened the measurement side of the paper. We added new sections to Appendix B. The first reports class-specific and platform-specific precision and recall, with bootstrap intervals, against the 171 of 200 validation items on which both human annotators agree (Table 3, p. 15). Accuracy against this consensus is 0.86. The two annotators agree with each other at $\kappa = 0.77$, and the model agrees with them at $\kappa = 0.61$ (annotator 1) and $\kappa = 0.74$ (annotator 2). The errors are not uniform: the model over-assigns each platform’s ideological minority, with precision of 0.64 for right-leaning labels on Bluesky and for left-leaning labels on Truth Social, against 0.98 and 0.93 for each platform’s largest class. Because the minority cells rest on few items, we test by bootstrap whether the gap is real (one-sided $p = 0.017$ on Bluesky and $p = 0.004$ on Truth Social) and report its size with its uncertainty.
 - The center category is now defined explicitly as a residual: it collects users who reach neither partisan threshold, which mixes moderate users, users who post about politics inconsistently, and users the classifier could not resolve (Section 3.5, p. 5). We interpret center-labeled users that way throughout, including in the motif results. To test whether this residual drives the result, we added an analysis that recomputes composition and alignment from partisan users only (Appendix B, p. 17). On the 28,083 cascades where this is possible, composition and alignment still account for 61 percent of the platform difference in breadth and 64 percent in depth, compared with 99 and 77 percent on the same cascades when center-labeled users are included. The share is smaller but still a majority.
 - We also added a label-noise analysis (Appendix B, pp. 15-17). It asks how much of the platform difference composition and alignment would still account for if the labels contained errors at the rates we measured. With the original labels, they account for 84 percent of the platform difference in breadth and 86 percent in depth. In each simulation draw, every user’s label is redrawn from the measured confusion matrix, composition and alignment are recomputed, and the model is refit. We ran 100 draws under each of four error processes: the measured platform-specific rates, a matrix pooled across platforms, a nested bootstrap of the validation sample, and the less favorable rates measured against a single annotator. In the median draw of each process, the share accounted for is between 70 and 86 percent, and in the least favorable of all 400 draws it is 66 percent. The pooled process lowers it the most; the platform-specific and bootstrap processes, which are closest to the error we measured, leave it near the original values. Label noise therefore lowers the share under some assumptions, but composition and alignment account for a majority of the platform difference in every simulation.
 - The results also hold when users are classified with thresholds of 0.5 and 0.7 in place of 0.6 (Table 2, p. 15).
- **SPC-4.** *Reframe explanatory claims. Replace causal language suggesting that ideology and*

alignment “explain” platform differences with more cautious language about association, unless additional analyses can support a stronger claim. Discuss the potential endogeneity between alignment ratio and cascade structure.

- We are grateful for this comment and agree. We have revised the manuscript throughout: composition and alignment are now described as associated with, or statistically accounting for, the platform difference in the abstract, introduction, results, and discussion. Where the submitted version said that platform differences disappear once ideology is modeled, the revision reports how much of the difference composition and alignment account for: roughly five sixths of the platform difference for both breadth and depth (Table 1, p. 8).
- On endogeneity: the alignment ratio is computed from the same reply edges that determine depth and breadth, so alignment and cascade structure are jointly realized features of one conversation. Cross-ideological exchanges may lengthen sequential discussion, and sustained discussion may in turn create more occasions for cross-ideological exchange, the possibility R3 raises with deep and star-like cascades. The Limitations section (Section 6, p. 10) states this and treats alignment as associated with cascade structure, and the Results point to it where Model 3c is introduced (p. 7).
- **SPC-5.** *Improve reporting of data and robustness. Add readable descriptive statistics on posts, cascades, cascade sizes, motif frequencies, user participation patterns, and topic model outputs. Include motif counts and explain how dependence among motifs is handled.*
 - Thank you for these suggestions and we think those details make the paper more comprehensive. We added a new appendix with Table 4 (Appendix C, p. 18), which reports, for each platform and cascade type, the number of cascades, the number of posts, and the number of cascades in each size range, as counts with percentages. The data cover May 30 to June 30, 2024 (Section 3.1, p. 4). Compiling this table surfaced something we had not reported: most reply cascades are a single post with no replies, and far more so on Bluesky (71.3 percent) than on Truth Social (43.3 percent). These single posts are identical in structure but not in frequency, and because they all sit at the same point of the scaling plot, how many of them each platform has affects the fitted lines. We therefore refit every model on the 47,670 cascades with more than one post and report the results in a new Table 6 (p. 21). The platform difference is smaller on this sample, and composition and alignment still account for 81 to 92 percent of it.
 - We added a new appendix (Appendix G, pp. 18-19) whose Table 5 (p. 19) gives the observed count of each of the 54 ideology-labeled three-node motifs on both platforms, next to the z-scores in Figures 4 and 5. On dependence: we count every three-node chain in a tree, including chains that share nodes, so a chain of four posts contains two overlapping three-node chains and both are counted. The randomized trees are counted the same way, so overlap raises the observed and the expected counts alike and does not bias the comparison. Overlap does make the scores of neighboring motifs correlated, so we read them together, as patterns across motif families. Appendix G also describes the null model: 100 randomized trees per platform, built by swapping reply targets within each cascade (5m attempts for a cascade of m edges), which keeps each cascade’s size and each participant’s ideology fixed and randomizes who replies to whom. The submitted main text gave the number of randomizations as 1,000,000, which did not match the code or the appendix; both now state 100.
 - User participation patterns are reported in the new Appendices E and F (see SPC-2). A new Appendix H (p. 19) reports the topic model: it was fit on 125,623 root

posts, BERTopic’s outlier reduction step reassigns low-density posts to their nearest topic, leaving 21 posts (0.02 percent) unassigned, and manual labeling yields the eleven categories used in Model 4.

- The SPC’s summary also asked about back-and-forth reply chains between the same two users and about figure readability; we address these under R1’s general comments 6 and 1.
- **SPC-6.** *Resolve methodological inconsistencies. Ensure that the regression estimator, loss function, and standard error procedure are described consistently across the main text and appendix.*
 - Thank you for catching this, and we apologize for the confusion; it was an error in the submitted version. The reply-cascade models are robust linear models with the Huber loss. The appendix paragraph describing linear regression with HC3 standard errors was left over from an earlier specification and has been removed. The only analysis estimated by OLS with HC3 standard errors is the new repost reconstruction check, where the outcomes take few distinct values and the Huber scale estimate is degenerate; the appendix now says so.
 - Checking this surfaced a related problem. Because 61 percent of reply cascades are a single post at the origin of the scaling plot, the default median-absolute-deviation scale estimate collapses to nearly zero, and the standard errors in the submitted Table 1 could not be interpreted. We refit every model with Huber’s proposal 2 scale, which does not degenerate here, and report standard errors from the sandwich covariance estimator (Section 3.6, p. 5; Table 1, p. 8). Point estimates shift, most visibly for Model 3c on breadth, from 0.023 to 0.071 against a baseline of 0.419, and composition and alignment still account for about five sixths of the platform difference for both outcomes. A new Table 6 (p. 21) reports the results under both scale estimators, with and without single-post cascades: the share accounted for ranges from 81 to 94 percent across the four combinations. The threshold, encoding, and label-noise analyses in the appendix use the same estimator as Table 1.
- **SPC-7.** *Sharpen implications and limitations. Explain why the observed structural differences matter, what they suggest about platformed political discussion, and how limited the conclusions are to the 2024 U.S. election context and Biden/Trump-related discourse.*
 - Thank you for encouraging us to spell this out. The Discussion (Section 5, pp. 9-10) now states what the two cascade shapes mean. Wide, shallow reply cascades consist of many users responding to the same post with little exchange among them; narrow, deep cascades consist of replies that themselves draw replies over several turns. The two shapes distribute attention differently, and here the difference arises between platforms with similar reply mechanics. The Discussion also states a methodological implication: comparisons based on activity or cascade size alone would record Truth Social as simply more active and miss the divergence, which appears only after conditioning on size.
 - The limitations are now a separate section (Section 6, p. 10). It states that the sample is candidate-centered and covers one month of the 2024 U.S. campaign; that the ideological composition we observe reflects that conversation and should not be read as the composition of either platform’s full user base; that both platforms are U.S.-centered, so the patterns describe a single national context; that our ideology labels are coarse positions within the sampled Biden and Trump discourse; and that the alignment result is an association. The conclusion the evidence supports is that platforms with similar

reply mechanics can exhibit different conversational structures conditional on cascade size, and that these differences are associated with who participates and how they engage across ideological lines.

Responses to Reviewer 1

Repost cascade reconstruction

- *You have reconstructed the cascades for reposts in a very deterministic manner according to section 3.2. I have no way of gauging whether this reconstruction is in any way realistic and I am concerned that the results which show that Bsky and Truth Social have similar report cascade structure are an artefact of this reconstruction process. I would want to see substantial validation and robustness checks to assess the influence of your choice of reconstruction mechanism on your results. The text refers to “empirical” cascades - if your cascades are reconstructed, then these are not empirical but modelled. I suggest either (1) cutting all the repost content and focusing on the reply trees, or (2) adding substantial robustness checks which assess the impact of the cascade reconstruction mechanism.*
 - Thank you for this detailed comment and for laying out the two options; we took the second. Because the concern is that one deterministic rule could produce the similarity, we re-derived every repost cascade under alternative deterministic rules and a random-rule ensemble, and found that the rule used in the paper gives the largest platform difference of any of them. Under all of them the platform difference for reposts stays close to zero, far below that for reply cascades (new Appendix D and Figure 9, pp. 17-18; details under SPC-1). We also removed “empirical” for these cascades, which are now called reconstructed throughout.

Ideology labels

- *A lot of the results rely on a comparison between different ideological groups (left, centre, right). This is interesting, but it is also extremely reliant on the method used to label the ideology of users. This concerns me for multiple reasons: (1) you seem to find vastly more left leaning users of Truth Social than the wider literature would suggest, and likewise far more right leaning users on Bsky than the literature would suggest (I am thinking of a Bsky paper by Alex Bovet’s team in Zurich which looked into this). This makes me question your ideological labelling and all the results which depend on this labelling, e.g., the ideology-based cascade trees. It is also important because the literature consistently suggests that most social media users are not politically engaged (although this is of course less likely to be the case in threads mentioning Biden or Trump). I would ask the authors to go to some effort to convincing the reader of the validity of their ideology labelling - currently I am not convinced.*
 - Thank you for pressing us on this, and for pointing to the Bluesky study by Bovet’s group (Quelle and Bovet 2025), which Appendix B now compares with our estimates (p. 17). Our estimates describe users who take part in the political conversations our keywords define, posts mentioning Biden or Trump during one campaign month, and they are not estimates for either platform’s full user base. As you note, most users are not politically engaged, and less political users rarely appear in these conversations, so the ideological mix we observe need not match platform-wide estimates. Our labels also summarize the stance each user expresses in these conversations, which can differ from

that user’s broader political position. The Limitations section now states both points (p. 10).

- Part of the gap is also measurement error, in the direction you suspected: the model over-assigns each platform’s ideological minority (SPC-3). We added a correction for this to Appendix B (p. 17), which adjusts the estimated shares using the measured confusion matrix. It lowers the right-leaning share on Bluesky from 24.2 to 17.9 percent and the left-leaning share on Truth Social from 11.5 to 1.9 percent. The remaining gap for Bluesky may reflect the candidate-centered sample and differences in measurement across studies.
- Since our sample is not the platform population, the relevant test is whether the labels are reliable enough for the comparison they are used in. The new analyses under SPC-3 address this directly: class-specific and platform-specific validation, a test of the direction of the error, simulated label noise at the measured rates, alternative thresholds, and a partisan-only specification. Across all of them, composition and alignment account for a majority of the platform difference.

Motifs, topics, and data

- *I like Figures 4 and 5, but I would want to know how many of each motif are actually present in your dataset. This is crucial information for actually assessing the validity of your results.*
 - Thank you for this suggestion. A new Table 5 (p. 19) gives the observed count of all 54 ideology-labeled motifs on both platforms, alongside the z-scores in Figures 4 and 5. The counts matter for interpretation in the way you anticipated: on Truth Social, for example, Left to Right bounce-back chains are strongly overrepresented relative to the null model while remaining rare in absolute terms, which the Discussion now notes (p. 9).
- *Related to point 2 - different motifs are of course not independent of each other. Does your randomisation strategy account for this. In a chain of length 4, there are two 3 motifs, but these are not independent. How does this effect your results?*
 - Thank you for this question. The randomization does account for it, in the sense that matters for the z-scores: we count every three-node chain, including chains that share nodes, in both the observed and the randomized trees, so a chain of four posts contributes two overlapping three-node chains to both counts alike and the overlap does not bias the comparison. It does make the scores of neighboring motifs correlated, so we read them together, as patterns across motif families, and say so in Appendix G (pp. 18-19; see also SPC-5).
- *A bit more info on the topic modelling would be useful. In my experience most Twitter-like posts are too short to get a useful topic label and BERTopic tends to throw 50% + of the posts into an outlier group.*
 - We appreciate this suggestion. A new Appendix H (p. 19) describes the topic model. It was fit on 125,623 root posts. HDBSCAN initially assigns low-density posts to an outlier class; we applied BERTopic’s outlier reduction step with its default settings, which reassigns each such post to its nearest topic, leaving 21 posts (0.02 percent) unassigned. Manual labeling of the resulting clusters yields the eleven categories used in Model 4. Topic accounts for little of the platform difference, so the uncertainty that reassignment introduces is unlikely to affect the main comparison.

- *I want far more info about the data. How many posts? Over how long? The CCDFs in Fig 2 are useful to a point, but I cannot read off how many cascades of size N there actually are. Can you provide more of this data in a readable format?*
 - A new Table 4 (Appendix C, p. 18) reports, for each platform and cascade type, the number of cascades, the number of posts, and the number of cascades in each size range with percentages. The data cover May 30 to June 30, 2024 (Section 3.1, p. 4). There are, for example, 79,348 reply cascades on Bluesky containing 199,413 posts, and 43,841 on Truth Social containing 1,369,497 posts. The table also led to an additional analysis excluding single-post cascades, described under SPC-5.

Further analyses

- *I may have missed this, but how many of the reply chains are conversations between two users, i.e., a back and forth between the same two accounts?*
 - Thank you for this question, which we had not examined before. We added this analysis to the paper as a new appendix (Appendix F, p. 18). For every reply we checked whether its author also wrote its grandparent post, an $A \rightarrow B \rightarrow A$ exchange. Among cascades of depth two or more, 68.8 percent on Bluesky and 70.0 percent on Truth Social contain at least one, and such exchanges account for 20.0 and 15.6 percent of reply edges. Two-account back-and-forth is common on both platforms to a similar degree, so it does not explain the depth difference.
- *I'll be honest, I don't fully follow the results about the alignment of users... Some clearer explanation would be useful.*
 - Thank you for this feedback; the section was not as clear as it should have been. We rewrote the definition in Section 3.5 (p. 5) and added a worked example. Each reply edge is coded aligned if both users share an ideological label and misaligned otherwise, and a cascade's alignment ratio is the share of aligned edges. In the example, a thread has four replies: two right-leaning users reply to a right-leaning root author, a left-leaning user replies to the root, and that reply draws a response from a right-leaning user. Three edges connect users who share a label and one crosses, so the ratio is 0.75. The measure describes the conversation: the same user contributes aligned edges in one thread and crossing edges in another.
- *The justification for only looking at influencers on TS is poor. The lack of a natural gap in the cascade distribution is not really a natural justification. What if you just take the top $n\%$ of accounts on both platforms?*
 - Thank you for pressing on this and for suggesting a concrete alternative. The original manuscript did not explain clearly why we made this choice, and we now state the reasoning explicitly in Appendix A (p. 14). It lies in the shape of the two follower distributions. On Truth Social, eight accounts are separated from the rest of the distribution: the largest has 56 times the followers of the platform's 99.9th-percentile account, and together the eight hold 16.4 percent of all followers in our data. On Bluesky the largest account sits five times above the 99.9th percentile, continuous with the rest of the distribution, and the top eight hold 6.1 percent. The eighth-largest Truth Social account has more followers than the largest Bluesky account.
 - A top- n percent rule would therefore not equalize the comparison: on Bluesky it would remove accounts continuous with the rest of the distribution, and on Truth Social any

n large enough to capture the separated accounts would also sweep in ordinary ones. Model 2 accordingly treats the eight Truth Social accounts as a separate group, and they account for only a small share of the platform difference (Table 1, p. 8).

- *I don't particularly like that this is another US centric paper, but that is not really reason to reject. Just a shame not to diversify the focus.*
 - Thank you for raising this point. We agree with the concern, and we also wish the empirical focus of social media research were less concentrated on the U.S. context. At the same time, the U.S. focus in this paper follows from the case itself. The 2024 U.S. presidential election was an unusually high-salience and dramatic political moment, and Truth Social and Bluesky provide a particularly useful comparison because their political identities and user communities are closely connected to that political context. We see this as an empirically valuable setting for studying candidate-centered political diffusion, while also recognizing that it is not a basis for broad cross-national generalization. We have revised the Limitations section accordingly to make the U.S.-centered scope explicit and to note the need for future comparative work across national contexts.

Figures, presentation, and citations

- *A lot of the figures need a bit of cleaning up / further explanation.*
 - Thank you for this feedback. We revised the figures. In the motif figures (Figures 4 and 5, pp. 9-10), the motif family labels had been drawn inside the panels on top of the data; they now sit in a header band above the axes, and the captions explain the groupings and the $|Z| < 3$ threshold used to gray out small scores. In the topic and partisanship distributions (Figures 1 and 7, pp. 6 and 15), value labels on the longest bars fell outside the axes; both figures are rebuilt so every label sits inside, and the partisanship figure now uses a shared axis. The partial-residual diagnostic (Figure 11, p. 22) was three separate images; it is now a single row of panels with a smoother against each predictor's linear term, which shows the curvature the appendix describes. All figures are rendered at 300 dpi.
- *Overall the presentation is good, but there is a fair bit of unexplained notation. Also, the figures are messy in places with labels overlapping parts of the content. It is not always clear what is being shown. The latex has paragraph indents in places where there shouldn't be any.*
 - Thank you for pointing these out. We reviewed the manuscript for undefined notation and define the terms of the statistical models where they first appear (Section 3.6, p. 5). The stray paragraph indents came from blank lines after displayed equations and have been removed. The figure changes are described in our response to your first general comment.
- *The citations are appropriate and comprehensive. However, the literature is quite old in the context of social media research. It would perhaps be worth updating the literature to some papers which are more recent on information spreading on social media.*
 - Thank you for this suggestion. We added recent work on information spreading, including DeVerna et al. (2024) on how the cascade inference problem distorts measured diffusion, González-Bailón et al. (2024) on the diffusion and reach of information on Facebook during the 2020 U.S. election, and Slaughter et al. (2025) on how Community Notes reduce the diffusion of false information. We also added recent methodological work on using automated and language-model labels in downstream analysis (Egami et

al. 2023; TeBlunthuis, Hase, and Chan 2024; Vallejo Vera and Driggers 2025). The Related Work section now cites 17 studies published since 2023, including work on Bluesky and Truth Social.

Responses to Reviewer 2

Causal interpretation

- *I feel that the main “ideology explains the platform gap” claim is too strong: The paper measures ideological composition and alignment from the same completed reply cascades whose depth and breadth it is trying to explain. That makes these variables partly endogenous to the outcome. Alignment is not clearly a prior cause of cascade shape; it may simply describe the cascade after it has already formed. I think the safer claim is that ideology and alignment are associated with the platform gap, not that they explain it causally.*
 - Thank you for this comment; we agree. Composition and alignment are now described as associated with, or statistically accounting for, the platform difference throughout, and the Limitations section discusses the endogeneity you describe: alignment and cascade structure are jointly realized features of one conversation, so their association does not establish a direction of influence (Section 6, p. 10). Details under SPC-4.

Cascade construction

- *The cascade construction is ambiguous for me. The paper says cascade nodes are users, but reply trees are naturally built from posts/replies. If the same user replies multiple times, collapsing posts into one user node can change depth, breadth, and motif structure. Can authors further clarify this?*
 - Thank you for pointing this out; the text was unclear on this point. Nodes are posts: a user who replies several times contributes one node per reply, each carrying that user’s label (Section 3.2, p. 4). Collapsing repeated replies into one user node leaves the depth difference in place but shrinks the breadth difference from 0.160 to -0.032 on the same threads, and we report both representations (Appendix E, p. 18). Details under SPC-2.
- *In addition, the paper infers repost paths because platform APIs do not directly provide true diffusion paths. That is reasonable, but the inference depends on assumptions about follower relationships and timing. This may be further justified.*
 - Thank you for this suggestion. This is a common concern raised by all reviewers. We now test these assumptions directly. All repost cascades are rebuilt under alternative linking rules and under simulated follower-timing error, and the repost result holds throughout (Appendix D, p. 17). Details under SPC-1.

Generalizability

- *The study covers one month and only Biden/Trump-related discussion during a highly unusual campaign period, which makes the generalizability a little limited.*
 - Thank you for this comment. We agree that this limits what the paper can claim. The new Limitations section (Section 6, p. 10) states that the sample is candidate-centered, covers one month of the 2024 campaign, and describes a single national context, and it states the narrower conclusion the evidence supports. Details under SPC-7.

Ideology validation

- *Another concern is about data size. The LLM-based stance validation uses only 200 human-labeled replies, and the paper does not report enough class-specific or platform-specific performance. The “center” category also appears to include genuinely centrist users, mixed users, and uncertain users. Given ideology is the paper’s main explanatory variable, label noise could directly affect the results.*
 - Thank you for these suggestions. We now report class-specific and platform-specific performance with bootstrap intervals (Table 3, p. 15), define the center category explicitly as a residual and test a partisan-only specification, and propagate the measured label error through the models. The minority-class cells in the 200-item sample are small, so we report their uncertainty, test the direction of the error by bootstrap, and show that the substantive result survives label noise at the measured rates. Details under SPC-3.

Regression estimator

- *In addition to the weaknesses listed above, I find that the methods describe robust regression with Huber loss, while the appendix says linear regression with HC3 robust standard errors. These are different estimators. I suggest authors double check this.*
 - Thank you for catching this, and we apologize for the inconsistency. The models use the Huber loss, and the HC3 paragraph has been removed. Checking this also led us to correct the scale estimator and refit Table 1. Details under SPC-6.

Responses to Reviewer 3

Treatment of ideological alignment

- *The main claim that ideological composition and interaction alignment jointly account for the platform-level divergence in reply cascade scaling is interesting, but this interpretation requires caution. The alignment ratio is computed from the reply edges that constitute the cascades. Therefore, while alignment ratio may affect cascade structure, it is also possible that cascade structure affects the alignment ratio. For example, deep cascades and star-like cascades may provide different opportunities for cross-ideological interaction. I therefore recommend weakening the causal interpretation of this result, or strengthening the discussion and analysis of this issue.*
 - Thank you for this comment and for the example of deep and star-like cascades, which helped us state the problem clearly. We weakened the causal interpretation throughout, and the Limitations section now discusses the two-way relationship you describe: cross-ideological exchanges may lengthen sequential discussion, and sustained discussion may create further occasions for such exchanges (Section 6, p. 10). Details under SPC-4.

Construction of repost cascades

- *The paper uses the follower network to infer repost cascades, but the timing of post collection and follower-network collection may affect the accuracy of this inference. For example, suppose user u follows user v and u reposts v ’s post. The paper appears to infer that information flowed from v to u . However, is it possible that u followed v only after reposting v ’s post? If u followed v after the repost occurred, then inferring information flow from v to u would not be*

appropriate. The paper should discuss the potential impact of this type of inference error. It should also consider whether such errors may differ systematically between Truth Social and Bluesky.

- Thank you for this careful example. It describes a real limitation exactly: the follower networks are snapshots without edge creation dates, so a follow edge can postdate the repost it is used to explain. The two platforms' follower networks were collected at the same time, so the delay between posting and the snapshot is similar for both, although we do not assume the resulting error is identical. To bound its effect, we deleted 5 to 30 percent of follow edges at random before reconstruction; across that range the platform interactions remain an order of magnitude smaller than those for reply cascades (Appendix D, p. 17). Section 3.2 (p. 4) now states the snapshot limitation. Details under SPC-1.

Definition of cascades

- *In reply cascades, the same user may appear multiple times in the same thread. It is unclear how the paper handles such cases. For example, does the paper allow the same user to appear as multiple nodes in a single cascade? Or are nodes defined strictly as users, with only a specific reply, such as the first or last appearance, counted for each user? These details should be clarified.*
 - Thank you for raising this. The same user can appear as multiple nodes: nodes are posts, and a user who replies several times contributes one node per reply (Section 3.2, p. 4). We also report results with each user collapsed to a single node at the shallowest depth at which they appear (Appendix E, p. 18). Details under SPC-2.

Implications and limitations

- *The finding that cascade structures differ across platforms is interesting. However, the practical and theoretical implications of this finding are not very clear. It is also unclear whether the results are specific to the 2024 U.S. election context or whether they can be generalized more broadly. The generalizability of the findings may be addressed in future work, but the paper should provide a more careful discussion of its implications and limitations.*
 - Thank you for this suggestion. We rewrote the Discussion to state what the two cascade shapes imply (Section 5, pp. 9-10) and moved the limitations, including the dependence on the 2024 U.S. campaign and the candidate-centered sample, into a separate section (Section 6, p. 10). Details under SPC-7.

We thank the reviewers and the SPC again for their careful and constructive feedback. We feel that this revision has made the project substantially stronger and more robust. The comments pushed us to clarify several parts of the analysis, add sensitivity checks that we had not originally included, and narrow some of the interpretations where appropriate. We very much appreciate the time the reviewers spent on the paper and the opportunity to revise it.